

environment: link loss · sensor fault · agent dropout · communication delay

A precursor risk

diagnostic signals $y_e(k)$

RQA / permutation entropy



split conformal prediction

$$P[\bar{r}_e > \rho_e] \leq \alpha_e$$



$$\bar{r}_e(\alpha_e)$$

$$S(\alpha, \rho)$$

admissible structures

$S(\alpha, \rho)$

B structure selection

candidates $E \cup V \cup U$

**downward-closed
independence system**



restricted graphic matroid

Kruskal (polynomial)



$$s^* = (E, V, U)$$

selected structure

s^*

C safe recovery control

distributed MPC + CBF

$\bar{r}_e \rightarrow$ chance constraint



$\mathcal{X}_f(s), \Omega(s)$ invariant sets



safe transition, dwell time τ_d

recursive feasibility + safety

with probability $1 - \alpha$

coverage $1 - \alpha$

exact, polynomial time

feasibility across a switch

realised structure and control outcome fed back to the next risk evaluation